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Functional Requirements

1. User signup and login (email or OAuth), multiple profiles per account, parental controls
2. Browse catalog by genre, language, and new releases
3. Search content by title, actor, or genre
4. Stream video with adaptive bitrate based on network speed
5. Resume playback from the last watched position
6. Content upload and transcoding into multiple resolutions (admin or studio side)
7. Personalized recommendations based on watch history and ratings
8. Track watch history per profile
9. Ratings and reviews (thumbs up or down, star rating)
10. Subscription plans, recurring billing, upgrade, downgrade, or cancel
11. Multi device support (web, mobile, smart TV) with synced state
12. Offline download support on mobile devices
13. Notifications for new episodes and renewal reminders

Non-Functional Requirements

Requirement	Detail
Availability	99.99 percent uptime
Low latency	Video starts in under 2 seconds, API responses under 200ms
Scalability	250 million plus users, tens of millions of concurrent streams at peak
Durability	No loss of video assets, user data, or billing records
Consistency	Eventual consistency for catalog and recommendations, strong consistency for billing
Global reach	Multi region deployment, content served via CDN close to the user
Security	Encrypted delivery (DRM), secure payments (PCI DSS), encryption at rest and in transit
Fault tolerance	No single point of failure, graceful quality degradation instead of failure
Elasticity	Auto scale during peak hours (evenings, weekends, new releases)

Capacity Estimation

Assumptions

- Registered users: 250 million
- Daily active users (DAU): 100 million
- Average watch time per DAU: 2 hours per day
- Peak concurrency: 20 percent of DAU = 20 million concurrent streams
- Average bitrate per stream: 5 Mbps
- Catalog size: about 15,000 titles

Bandwidth

Peak bandwidth = 20,000,000 streams x 5 Mbps = 100 Tbps (served mostly via CDN edge caches)

Storage

15,000 titles x 10 renditions x 7 GB average = about 1.05 PB; x3 replication = about 3 PB total

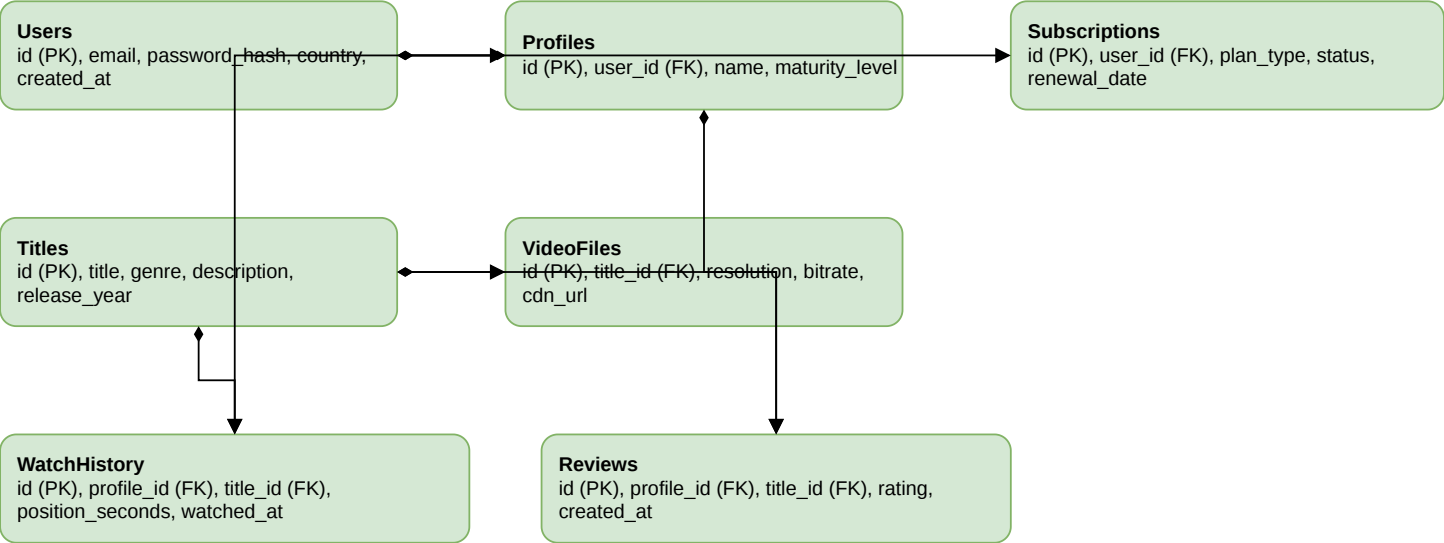
Request Rate (API layer)

100M DAU x 50 calls per day = 5 billion requests per day, average about 58,000 RPS, peak 175,000 to 290,000 RPS

Database Size

- Users table: 250M rows x 1 KB = about 250 GB
- Watch history: about 36.5 TB per year (time series, needs partitioning and TTL)
- Catalog metadata: 15,000 titles x 50 KB = about 750 MB

Database Structure (simplified ER model)



API Methods and Endpoints

Auth Service

POST	/api/v1/auth/signup
POST	/api/v1/auth/login
POST	/api/v1/auth/logout
POST	/api/v1/auth/refresh-token

User and Profile Service

GET	/api/v1/users/{userId}
PUT	/api/v1/users/{userId}
POST	/api/v1/users/{userId}/profiles
GET	/api/v1/users/{userId}/profiles
DELETE	/api/v1/users/{userId}/profiles/{profileId}

Catalog Service

GET	/api/v1/titles
GET	/api/v1/titles/{titleId}
GET	/api/v1/search?q={query}

Streaming Service

GET	/api/v1/titles/{titleId}/manifest
POST	/api/v1/titles/{titleId}/playback-session

Watch History Service

POST	/api/v1/profiles/{profileId}/history
GET	/api/v1/profiles/{profileId}/history

Recommendation Service

GET	/api/v1/profiles/{profileId}/recommendations
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Review Service

POST	/api/v1/titles/{titleId}/reviews
GET	/api/v1/titles/{titleId}/reviews

Billing Service

POST	/api/v1/subscriptions
GET	/api/v1/subscriptions/{subId}
PUT	/api/v1/subscriptions/{subId}
DELETE	/api/v1/subscriptions/{subId}

High-Level Architecture

